

Lecture #4

Inverse Trigonometric Functions

$\sin^{-1} \frac{1}{2}$ is the angle that has a sine value $\frac{1}{2}$

, then $\sin^{-1} \frac{1}{2} = 30^\circ = \frac{\pi}{6}$. Similarly

$$\cos^{-1} \frac{1}{2} = 60^\circ = \frac{\pi}{3}$$

$$\cos^{-1} 0.4226 \approx 65^\circ$$

$$\tan^{-1} 1 = 45^\circ = \frac{\pi}{4}$$

$$\tan^{-1} \sqrt{3} = 60^\circ = \frac{\pi}{3}$$

★ $\sin^{-1} 0 = 0, \pi, 2\pi, \dots$

$$\sec^{-1} 2 = \cos^{-1} \frac{1}{2} = 60^\circ = \frac{\pi}{3}$$

$$\operatorname{cosec}^{-1} \sqrt{2} = \sin^{-1} \frac{1}{\sqrt{2}} = 45^\circ = \frac{\pi}{4}$$

$$\cot^{-1} \frac{1}{\sqrt{3}} = \tan^{-1} \sqrt{3} = 60^\circ = \frac{\pi}{3}$$

Notice that $\left(\sec^{-1} \alpha = \cos^{-1} \frac{1}{\alpha} \right), \sec^{-1} \alpha \neq \frac{1}{\cos^{-1} \alpha}$

similarly

$$\operatorname{cosec}^{-1} \alpha = \sin^{-1} \frac{1}{\alpha}$$

$$\cot^{-1} \frac{\alpha}{\beta} = \tan^{-1} \frac{\beta}{\alpha}$$

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Remark:

$$\sin^{-1} 3$$

$$\cos^{-1}(-2.1)$$

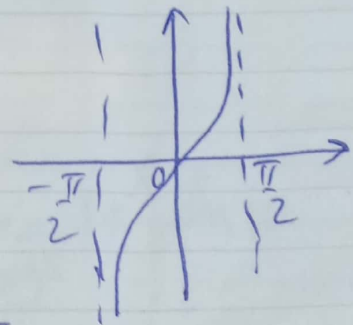
$$\sec^{-1} 0.3$$

$$\operatorname{cosec}^{-1}(-0.5)$$

ALL are
undefined

$$\lim_{x \rightarrow \infty} \tan^{-1} x = \tan^{-1} \infty = \frac{\pi}{2}$$

$$\lim_{x \rightarrow -\infty} \tan^{-1} x = \tan^{-1}(-\infty) = -\frac{\pi}{2}$$



Note:

$$\sin^{-1} x = y \rightarrow x = \sin y$$



$$\tan^{-1} n = \theta \rightarrow n = \tan \theta$$

$$u = \cos v \rightarrow v = \cos^{-1} u$$

$$s = \sec t \rightarrow t = \sec^{-1} s$$

$$\sin \sin^{-1} x = x$$

$$\tan \tan^{-1} y = y$$

$$\operatorname{cosec}^{-1} \operatorname{cosec} \alpha = \alpha$$

$$\cos^{-1} \cos \frac{\pi}{8} = \frac{\pi}{8}$$

and so on.

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Derivatives of inverse trig. functions:

$$y = \sin^{-1} x \rightarrow y' = \frac{1}{\sqrt{1-x^2}}$$

Proof: $y = \sin^{-1} x \rightarrow x = \sin y$

$$\frac{d}{dx} \Rightarrow 1 = \cos y \cdot y'$$

$$\Rightarrow y' = \frac{1}{\cos y} = \frac{1}{\sqrt{1-\sin^2 y}} = \frac{1}{\sqrt{1-x^2}}$$

Similarly, we can show

$$y = \cos^{-1} x \rightarrow y' = \frac{-1}{\sqrt{1-x^2}}$$

Examples: $y = \sin^{-1} x^3 \rightarrow y' = \frac{1}{\sqrt{1-(x^3)^2}} (3x^2)$

$$y = \cos^{-1} 10x \rightarrow y' = \frac{-1}{\sqrt{1-(10x)^2}} (10)$$

③

Example: Find $y'(x)$

a) $y = \sin^{-1} e^x$

b) $y = \cos^{-1} 10^x$

c) $y = \sin^{-1} \sqrt{x}$

d) $y = \cos^{-1} \frac{1}{x}$

e) $y = \sqrt{\sin^{-1} x}$

Solution:

a) $y' = \frac{1}{\sqrt{1-(e^x)^2}} e^x$

b) $y' = \frac{-1}{\sqrt{1-(10^x)^2}} 10^x \ln 10$

c) $y' = \frac{1}{\sqrt{1-(\sqrt{x})^2}} \cdot \frac{1}{2\sqrt{x}}$

d) $y' = \frac{-1}{\sqrt{1-(\frac{1}{x})^2}} \left(-\frac{1}{x^2}\right)$

e) $y = \frac{1}{2} (\sin^{-1} x)^{-\frac{1}{2}} \cdot \frac{1}{\sqrt{1-x^2}}$

Notes

$$\frac{d}{dx} \sqrt{x} = \frac{1}{2\sqrt{x}}$$

$$\frac{d}{dx} \frac{1}{x} = \frac{-1}{x^2}$$

Rule:

$$\frac{d}{dx} \tan^{-1} x = \frac{1}{1+x^2}$$

$$\frac{d}{dx} \cot^{-1} x = \frac{-1}{1+x^2}$$

Proof: $y = \tan^{-1} x \rightarrow x = \tan y$

$$\frac{d}{dx} \Rightarrow$$

$$1 = \sec^2 y \cdot y'$$

$$\text{then } y' = \frac{1}{\sec^2 y} = \frac{1}{1+\tan^2 y}$$

$$= \frac{1}{1+x^2}$$

$$\text{then } y' = \frac{1}{1+x^2}$$

Similarly, $\frac{d}{dx} \cot^{-1} x = \frac{-1}{1+x^2}$

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Examples Find $y'(x)$

a) $y = \tan^{-1} x^{10}$

b) $y = \cot^{-1}(\ln x)$

c) $y = \tan^{-1}(x^\pi)$

d) $y = \cot^{-1}(x^e)$

e) $y = e^{\tan^{-1} x}$

f) $y = e^{\cot^{-1} 4x}$

Solution:

a) $y' = \frac{1}{1 + (x^{10})^2} (10x^9)$

b) $y' = \frac{-1}{1 + (\ln x)^2} \cdot \frac{1}{x}$

c) $y' = \frac{1}{1 + (x^\pi)^2} \pi x^{\pi-1}$

d) $y' = \frac{-1}{1 + (x^e)^2} e x^{e-1}$

e) $y' = e^{\tan^{-1} x} \cdot \frac{1}{1+x^2}$

~~f) $y' = \frac{-1}{1 + (\cot^{-1} 4x)^2} \cdot \frac{-1}{1+(4x)^2} (4)$~~

f) $y' = e^{\cot^{-1} 4x} \left(\frac{-1}{1+(4x)^2} \right) (4)$

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Rule:

$$\frac{d}{dx} \sec^{-1} x = \frac{1}{x\sqrt{x^2-1}}$$

$$\frac{d}{dx} \operatorname{cosec}^{-1} x = \frac{-1}{x\sqrt{x^2-1}}$$

proof: $y = \sec^{-1} x \rightarrow x = \sec y$

$$\frac{d}{dx} \Rightarrow$$

$$1 = \sec y \tan y \, y'$$

$$y' = \frac{1}{\sec y \tan y} = \frac{1}{\sec y \sqrt{\sec^2 y - 1}}$$

Note: $\sec^2 y = 1 + \tan^2 y$

$$\Rightarrow \tan y = \sqrt{\sec^2 y - 1}$$

then $y' = \frac{1}{\sec y \sqrt{\sec^2 y - 1}} = \frac{1}{x\sqrt{x^2-1}}$

similarly $\frac{d}{dx} \operatorname{cosec}^{-1} x = \frac{-1}{x\sqrt{x^2-1}}$

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Examples Find $y'(x)$

(a) $y = \sec^{-1} \sqrt{x}$

(b) $y = \operatorname{cosec}^{-1} \frac{1}{x}$

(c) $y = e^{\sec^{-1} x}$

(d) $y = e^{\sec^{-1} 10x}$

(f) $y = \ln(\operatorname{cosec}^{-1} x)$

(g) $y = \ln(\operatorname{cosec}^{-1} x^4)$

Solution:

(a) $y' = \frac{1}{\sqrt{x} \sqrt{(\sqrt{x})^2 - 1}} \cdot \frac{1}{2\sqrt{x}}$

(b) $y' = \frac{-1}{\frac{1}{x} \sqrt{(\frac{1}{x})^2 - 1}} \left(-\frac{1}{x^2} \right)$

(c) $y' = e^{\sec^{-1} x} \cdot \frac{1}{x \sqrt{x^2 - 1}}$

(d) $y' = e^{\sec^{-1} 10x} \cdot \frac{1}{(10x) \sqrt{(10x)^2 - 1}} (10)$

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$$\textcircled{e} \quad y = \frac{1}{\operatorname{cosec}^{-1} x} \left(\frac{-1}{x\sqrt{x^2-1}} \right)$$

$$\textcircled{f} \quad y = \frac{1}{\operatorname{cosec}^{-1} x^4} \cdot \frac{-1}{x^4 \sqrt{(x^4)^2 - 1}} (4x^3)$$

In summary:

$y(x)$	$y'(x)$
$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}$
$\cos^{-1} x$	$\frac{-1}{\sqrt{1-x^2}}$
$\tan^{-1} x$	$\frac{1}{x^2+1}$
$\cot^{-1} x$	$\frac{-1}{x^2+1}$
$\sec^{-1} x$	$\frac{1}{x\sqrt{x^2-1}}$
$\operatorname{cosec}^{-1} x$	$\frac{-1}{x\sqrt{x^2-1}}$

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Example: Find $y'(x)$

a) $y = \sqrt{\cos^{-1} e^x}$

b) $y = 3^{\sin^{-1} \frac{1}{x}}$

★ c) $y = \tan \cot^{-1} x$

d) $y = e^{\tan^{-1}(x\sqrt{x})}$

e) $y = \sec^{-1} \frac{x}{1+x}$

f) $y = 4^{\operatorname{cosec}^{-1} \sqrt{x}}$

solution:

a) $y' = \frac{1}{2} (\cos^{-1} e^x)^{-\frac{1}{2}} \left(\frac{-1}{\sqrt{1-(e^x)^2}} \right) e^x$

b) $y' = 3^{\sin^{-1} \frac{1}{x}} (\ln 3) \left(\frac{1}{\sqrt{1-(\frac{1}{x})^2}} \right) \left(-\frac{1}{x^2} \right)$

c) $y' = \sec^2(\cot^{-1} x) \left(\frac{-1}{1+x^2} \right)$

d) $y' = e^{\tan^{-1}(x\sqrt{x})} \left(\frac{1}{1+(x\sqrt{x})^2} \right) \left(\frac{3}{2} \sqrt{x} \right)$

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$$e) y' = \frac{1}{\left(\frac{x}{1+x}\right) \sqrt{\left(\frac{x}{1+x}\right)^2 - 1}} \cdot \frac{(1+x) - x}{(1+x)^2}$$

$$f) y' = \frac{\operatorname{cosec}^{-1} \sqrt{x}}{4} \cdot \frac{(-1)}{(\ln 4) \left(\sqrt{x} \sqrt{(\sqrt{x})^2 - 1} \right)} \cdot \frac{1}{2\sqrt{x}}$$

Examples $y = \sec^{-1} \sec(\sqrt{x^2 + 1})$

find y'

solution

$$y = \sec^{-1} \sec(\sqrt{x^2 + 1})$$

$$y = \sqrt{x^2 + 1} = (x^2 + 1)^{1/2}$$

$$y' = \frac{1}{2} (x^2 + 1)^{-1/2} (2x)$$

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Notice that:

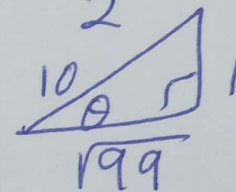
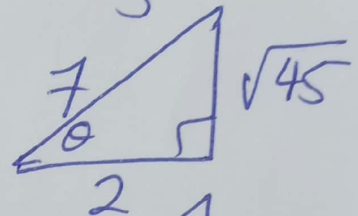
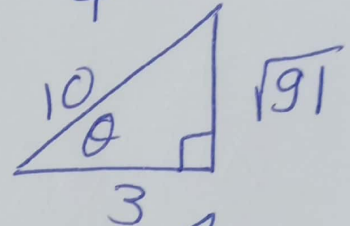
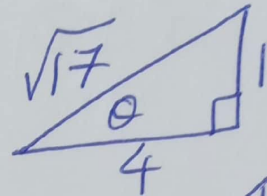
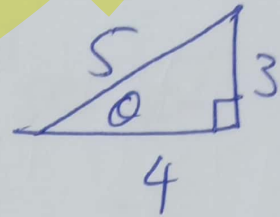
$$\sin \tan^{-1} \frac{3}{4} = \frac{3}{5}$$

$$\cos \cot^{-1} 4 = \frac{4}{\sqrt{17}}$$

$$\tan \cos^{-1} 0.3 = \frac{\sqrt{91}}{3}$$

$$\sin \sec^{-1} \frac{7}{2} = \frac{\sqrt{45}}{7}$$

$$\tan \operatorname{cosec}^{-1} 10 = \frac{1}{\sqrt{99}}$$



Also:

$$\tan \cot^{-1} 10 = \frac{1}{10}$$

$$\sec \cos^{-1} \frac{2}{7} = \frac{7}{2}$$

$$\operatorname{cosec} \sin^{-1} \frac{1}{100} = 100$$

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Example: Evaluate in terms of α

a) $\tan \cot^{-1} \alpha$

b) $\sec^{-1} \sec \sqrt{\alpha}$

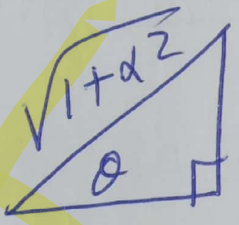
c) $\sin \tan^{-1} \alpha$

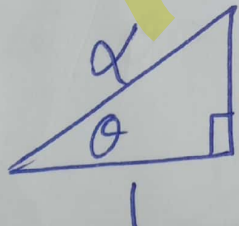
d) $\cot \sec^{-1} \alpha$

Solutions:

a) $\tan \cot^{-1} \alpha = \frac{1}{\alpha}$

b) $\sec^{-1} \sec \sqrt{\alpha} = \sqrt{\alpha}$

c)  $\alpha \Rightarrow \sin \tan^{-1} \alpha = \frac{\alpha}{\sqrt{1+\alpha^2}}$

d)  $\sqrt{\alpha^2-1} \Rightarrow \sin \tan^{-1} \alpha = \frac{\sqrt{\alpha^2-1}}{\alpha}$

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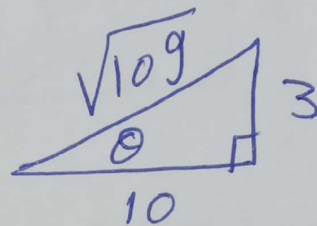
Example: Without using a calculator, find

a) $\sin(2 \tan^{-1} 0.3)$

b) $\sin(\cos^{-1} 0.2 + \sec^{-1} 4)$

Solution:

a) let $\theta = \tan^{-1} 0.3$

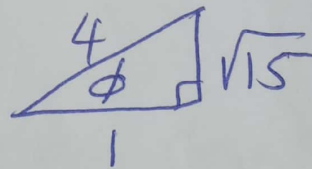
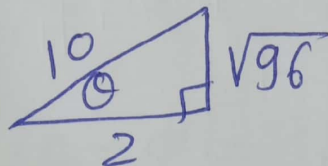


then $\boxed{\sin(2\theta) = 2 \sin \theta \cos \theta}$

$$\Rightarrow \sin(2\theta) = 2 \left(\frac{3}{\sqrt{109}} \right) \left(\frac{10}{\sqrt{109}} \right)$$

b) let $\theta = \cos^{-1} 0.2$

$\phi = \sec^{-1} 4$



then $\sin(\theta + \phi) = \sin \theta \cos \phi + \cos \theta \sin \phi$
 $= \left(\frac{\sqrt{96}}{10} \right) \left(\frac{1}{4} \right) + \left(\frac{2}{10} \right) \left(\frac{\sqrt{15}}{4} \right)$

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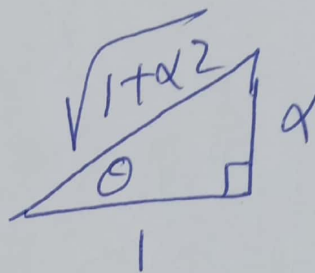
Example! Find in terms of α, β

a) $\sin(2 \tan^{-1} \alpha)$

b) $\sin(\cos^{-1} \alpha + \operatorname{cosec}^{-1} \beta)$

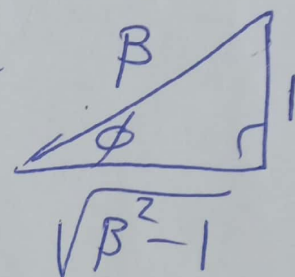
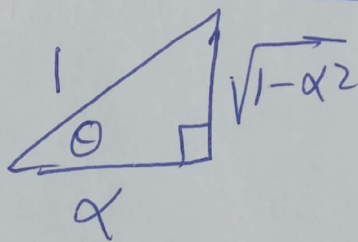
Solution:

let $\theta = \tan^{-1} \alpha$



$$\begin{aligned} \text{then } \sin 2\theta &= 2 \sin \theta \cos \theta = 2 \left(\frac{\alpha}{\sqrt{1+\alpha^2}} \right) \left(\frac{1}{\sqrt{1+\alpha^2}} \right) \\ &= \frac{2\alpha}{1+\alpha^2} \end{aligned}$$

b) let $\theta = \cos^{-1} \alpha$
 $\phi = \operatorname{cosec}^{-1} \beta$



then $\sin(\theta + \phi) = \sin \theta \cos \phi + \cos \theta \sin \phi$

$$= \left(\frac{\sqrt{1-\alpha^2}}{1} \right) \left(\frac{\sqrt{\beta^2-1}}{\beta} \right) + (\alpha) \left(\frac{1}{\beta} \right)$$

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